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Bounds and structure of the cyclotomic shortness of Φ_1^N

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To my parents,
For their unconditional support.

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Declaration on the use of Generative AI

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Abstract

This master's thesis looks at *cyclotomic shortness*: the idea of reducing the sum of a polynomial's absolute coefficients by multiplying it by n-th cyclotomics. The main focus is on the polynomial $(z-1)^N$. We find a lower bound using basic constraints on the roots, and derive upper bounds by testing different candidate polynomials. We also use the uncertainty principle on finite groups to prove structural results about the *short polynomial*. Finally, we present several smaller findings that help sharpen our intuition for finding these short polynomials.

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1 Introduction

Following a discussion regarding the direction of my Master's thesis, Professor Maryna Viazovska recommended the book *Around the Unit Circle: Mahler Measure, Integer Matrices and Roots of Unity* by McKee and Smyth [14]. This recommendation proved to be pivotal. While reading, I was immediately drawn to one of the most fundamental objects in number theory and algebra: Cyclotomic polynomials.

Cyclotomic polynomials $\Phi_n(z)$ stand as central objects in modern mathematics. They act as the minimal polynomials of the primitive n-th roots of unity, bridging the arithmetic of integers with the geometry of the circle. In the realm of Algebra, they provide the concrete realization of abelian Galois extensions over $\mathbb{Q}$ [13]. Their significance is equally profound in Number Theory, notably appearing in the foundational work of Gauss [12] and underpinning critical results regarding prime numbers and factorization [4, 1]. The formal definition and some of the basic properties of cyclotomic polynomials are detailed in Section 2.

While the general properties of these polynomials sparked my interest, I eventually focused on a specific metric discussed early in the book: the notion of *shortness*. The *length* of a polynomial P is defined as the sum of the absolute values of its coefficients. The *shortness* of P, denoted $sh(P)$, is the minimal length achievable by multiplying P by a cyclotomic polynomial. A more thorough explanation of these notions is provided in Section 3.

Why is shortness useful? Beyond its intrinsic combinatorial interest, shortness plays a crucial role in finding compact representations of algebraic numbers. As noted by El-Serafy and McKee, if we wish to preserve the Mahler measure of a polynomial while reducing its complexity, we look for multiples with small length [9]. Indeed, for many known small Mahler measures, the minimal polynomial has a multiple with a very small length.

However, the calculation of shortness depends heavily on the location of the roots. If a polynomial has a *house* (the maximum modulus of its roots) strictly greater than $1 + \epsilon$, the number of polynomials with a given shortness is finite, and the problem is effectively soluble [9]. This relies on results by Dobrowolski, who showed that one can in principle compute the shortness for such polynomials, although the process remains computationally expensive [8]. This solubility will be discussed further in Section 3.

This thesis, however, tackles a different beast. We focus on the pure cyclotomic case, specifically the polynomial $\Phi_1^N(z) = (z-1)^N$. Here, all roots lie perfectly on the unit circle, meaning the house is exactly 1. Consequently, the standard bounds and algorithms provided by El-Serafy and McKee or Dobrowolski do not apply directly. This places us in a unique context where many established tools cannot be used.

The specific problem I chose to tackle stems from Exercise 1.37 in *Around the Unit Circle* [14]:

> Let P be a cyclotomic polynomial and let N be the largest multiplicity of any zero of P. Show that $sh(P) \leq sh((z-1)^N)$.

This result (proven in Section 3) implies that by analyzing the shortness of $(z-1)^N$, one can effectively bound the shortness of any cyclotomic polynomial. This drove me to investigate the open problem of determining $sh((z-1)^N)$ for all N. While McKee and Smyth's book serves as the foundation and inspiration for this work [14], the specific constraints of the house = 1 case require us to explore methods distinct from those typically used for small Mahler measures.

Ideally, if we could restrict the search space of candidate polynomials, for instance, by establishing an upper bound on their degree, the calculation of shortness would become a finite problem (albeit computationally heavy). However, the specific nature of our context prevents such a straightforward approach. As we will discuss in Section 3, the shortness of a cyclotomic polynomial remains invariant under multiplication by any n-th cyclotomic polynomial with sufficiently low multiplicity. Therefore we cannot simply impose a global degree bound to make the search finite. Given these constraints, our approach shifts from a direct search to a theoretical analysis of bounds and structure.

We begin in Section 3 by formally defining the concepts of length and cyclotomic shortness, exploring the consequences of the result mentioned above and explaining why naive degree bounding is not a feasible option. We then tackle the bounds of the shortness of Φ_1^N in Section 4, where we establish theoretical bounds, while also providing exact values for small N to build intuition. In Section 5, we introduce an uncertainty principle for finite abelian groups to analyze the support of the coefficients from a structural perspective, offering new insights into the constraints governing these polynomials. In a detour to classical partition theory, we also examine the truncated Euler function $\prod_{k=1}^{N}(1-z^k)$, inspired by the Pentagonal Number Theorem, as a candidate for an upper bound in Section 6.

Notation and Definitions

To conclude this introduction, we briefly recall three standard definitions to fix the notation and terminology used throughout this thesis.

Definition 1.1. For a positive integer n, Euler's *totient function* $\varphi(n)$ counts the number of positive integers less than or equal to n that are relatively prime to n.

Definition 1.2. The *weight* of a polynomial $P(z) = \sum a_i z^i$, denoted by $w(P)$, is defined as the number of its non-zero coefficients.

Definition 1.3. Following the convention of McKee [14], we define the reciprocal of a polynomial $P(z)$ of degree d as $P^*(z) = z^d P(1/z)$. A polynomial P is said to be *reciprocal* if $P^*(z) = c_P P(z)$ for some constant c_P.

2 Basics of Cyclotomics

We start by defining the main object of this theme: *cyclotomic* polynomials. This definition relies on the notion of the *n-th cyclotomic*, which will be firstly precisely defined using Galois theory and then redefined in a more practicable way for our explicit setup. The objective is to make some of the main properties of this object more clear by taking the more general path.

Some freedom is taken in the choice of definitions. For instance in proposition 2.8 we define directly $\omega_n = e^{2\pi i/2} \in \mathbb{C}$ while in the main reference [14], it is defined as any of the n-th primitive roots of unity. However the results of this section should make it clear why the definition are actually equivalent. We work on any field K.

Definition 2.1. Let $n \in \mathbb{N}^*$ and define:

$$\mu_n(K) = \{\rho \in K \mid \rho^n = 1\}$$

Note that this set inherit a group structure from $K^\times$. The elements of $\mu_n(K)$ are called the *n-th roots of unity* in K. An element $\omega \in \mu_n(K)$ is called *primitive* if it is a generator of the whole group.

Lemma 2.2. *$\mu_n(K)$ is a finite cyclic group.*

Proof. Note that as $\mu_n(K) \subset K^\times$, it inherits an abelian subgroup structure. Furthermore as the number of roots of $\rho^n - 1$ is at most n, such group is finite. By the structure of finite abelian group (see theorem 2.1. of [13]), there exists integers $d_1|d_2|\dots|d_r$ (with $d_i > 1$) such that $\mathrm{card}(\mu_n(K)) = \Pi_i d_i$ and:

$$\mu_n(K) \cong C_{d_1} \times C_{d_2} \times \cdots \times C_{d_r}$$

where C_d denotes the cyclic group of order d. The number of elements in $\mu_n(K)$ satisfying $\rho^{d_r} - 1 = 0$ is given by:

$$\Pi_i \gcd(d_i, d_r) = \Pi_i d_i$$

where the second equality follow from the hypothesis $d_1|d_2|\dots|d_r$. This immediately forces $r = 1$, indeed if $r > 1$ then $\Pi_i d_i \geq 2d_r > d_r$ contradicting the fact that $\rho^{d_r} - 1$ has at most d_r roots. ∎

Remark 2.3. If $\mathrm{card}(\mu_n(K)) = n$, we shall call a primitive element a *primitive n-th roots of unity*. Furthermore, note that by the last lemma, if $\mathrm{card}(\mu_n(K)) = n$, then $\mu_n(K) \cong \mathbb{Z}_n$ and one can easily compute the number of primitive n-th roots of unity, i.e. the number of generators of $\mathbb{Z}_n$, as it is given by $\varphi(n)$.

Remark 2.4. Let's consider $L|K$ a splitting field for $z^n - 1 \in \mathbb{K}[z]$. Note that in the case where $\gcd(n, \mathrm{char}(K)) = 1$, the polynomial $z^n - 1$ has no multiple roots as $\gcd(z^n - 1, nz^{n-1}) = 1$. In such case $z^n - 1$ becomes separable over K. As a splitting field for a separable polynomial is equivalent to being Galois (see theorem 3.18. of [24]), the field L is Galois and $\mu_L(n)$ has cardinality n. Also note that primitive elements becomes primitive n-th roots of unity in such case.

Definition 2.5. Set $n \in \mathbb{N}^*$ and a field K such that $\gcd(n, \mathrm{char}(K)) = 1$. Let L be a splitting field for $z^n - 1 \in K[z]$. One can define the *n-th cyclotomic* as:

$$\Phi_{n,K}(z) = \prod_{\substack{\omega \in \mu_n(L) \\ \omega \text{ primitive}}} (z - \omega)$$

Remark 2.6. Note that by remarks 2.3 and 2.4 we have $\deg(\Phi_{n,K}) = \varphi(n)$.

One of the most fundamental result about n-th cyclotomic polynomial is the the fact that their coefficients lie in K. The proof here rely on the fact that the extension L is actually a Galois extension.

Lemma 2.7. *The polynomial $\Phi_{n,K}(z)$ has coefficient in K.*

Proof. The coefficients are symmetric functions in the primitive n-th roots. As the roots are permuted by $\mathrm{Gal}(L|K)$, the coefficients are invariant under $\mathrm{Gal}(L|K)$ and thus lies in K. ■

We are now ready to introduce the n-th cyclotomic in $\mathbb{Z}[z]$, the main object of our study. We'll drop the general case and specialize in $\mathbb{Q}$.

Proposition 2.8. *For $K = \mathbb{Q}$ and defining $\omega_n = e^{2\pi i/n} \in \mathbb{C}$, definition 2.5 becomes:*

$$\Phi_{n,\mathbb{Q}}(z) = \prod_{\substack{1 \le j \le n \\ \gcd(j,n)=1}} (z - \omega_n^j)$$

Proof. Set L a splitting field for $z^n - 1 \in \mathbb{Q}[z]$. Using the fundamental theorem of Algebra, one can embed L into $\mathbb{C}$. This allows us to represent the roots explicitly using the complex exponential function. Furthermore, $\mathbb{Q}$ has zero characteristic, ensuring that $\mu_n(L)$ is a cyclic group of n elements. It is clear that $\omega_n \in \mu_n(L)$ and as ω_n^j generates n distinct elements of $\mathbb{C}$ for $j = 1, \ldots, n-1$, one can conclude that ω_n is a primitive element. To conclude the equivalence of the definitions, the last thing to verify is the following: ω is primitive if and only if $\omega = \omega_n^j$ for some $j = 1, \ldots, n$ with $\gcd(j, n) = 1$. Let $\omega = \omega_n^j$. The order of an element ω_n^j is given by the formula:

$$\mathrm{ord}(\omega_n^j) = \frac{n}{\gcd(j, n)}.$$

By definition, ω is a primitive n-th root of unity if and only if its order is exactly n. Therefore, ω is primitive equivalent to $\gcd(j, n) = 1$, concluding the proof. ■

Remark 2.9. To make notations simpler, the n-th cyclotomic polynomials described on $K = \mathbb{Q}$ will be denoted as: $\Phi_n := \Phi_{n,\mathbb{Q}}$. From now on, the term *n-th cyclotomic polynomial* will refer specifically to Φ_n.

This method of introducing n-th cyclotomic polynomials may appear somewhat formal, and indeed it is. However, the literature rarely explicitly links the general definition 2.5 with the more concrete one in Proposition 2.8. Even though the link is direct, it is important to emphasize it. Furthermore, we will see in the next section that the coefficients actually lie in $\mathbb{Z}$. This is such a remarkable fact that the chosen approach aims to provide more intuition as to why, especially with Lemma 2.7, which uses Galois theory to show that the coefficients are in K and not in the extension.

2.1 Properties of the n-th cyclotomic

Our starting point is the n-th cyclotomic decomposition of $z^n - 1$ and ultimately establish the integrality of the coefficients of Φ_N.

Lemma 2.10. *For all $n \in \mathbb{N}^*$, we have:*

$$z^n - 1 = \prod_{d|n} \Phi_d(z)$$

Proof. The roots of the polynomial $z^n - 1$ are precisely the n-th roots of unity forming the set μ_n. We partition this set according to the multiplicative order d of each element. Since the order of an element must divide n, we can write μ_n as a disjoint union:

$$\mu_n = \bigsqcup_{d|n} P_d, \quad \text{where } P_d = \{\omega \in \mu_n \mid \operatorname{ord}(\omega) = d\}$$

By definition, the roots of $\Phi_d(z)$ are exactly the elements of P_d (the primitive d-th roots of unity). Thus, translating this partition into polynomials yields:

$$z^n - 1 = \prod_{\omega \in \mu_n} (z - \omega) = \prod_{d|n} \left(\prod_{\omega \in P_d} (z - \omega) \right) = \prod_{d|n} \Phi_d(z)$$

■

Corollary 2.11.

$$\Phi_n(z) = \prod_{d|n} (z^{n/d} - 1)^{\mu(d)} = \prod_{d|n} (z^d - 1)^{\mu(n/d)}$$

where μ is the Möbius function.

Proof. Immediate consequence of lemma 2.10 and using Möbius inversion formula. ■

Remark 2.12. In particular this showcases that for any prime p, we have:

$$\Phi_p(z) = \frac{z^p - 1}{z - 1} = 1 + z + \cdots + z^{p-1}$$

Theorem 2.13. *The coefficients of Φ_n lie on $\mathbb{Z}$*

Proof. For $n = 1$, $\Phi_1(z) = z - 1 \in \mathbb{Z}[z]$ is clear. Let's work by induction. Assume that for all $k < n$, $\Phi_k(z) \in \mathbb{Z}[z]$. Using theorem 2.10, we have:

$$z^n - 1 = \prod_{d|n} \Phi_d(z) = \Phi_n(z) \cdot \prod_{\substack{d|n \\ d<n}} \Phi_d(z)$$

Denote $Q(z) = \prod_{\substack{d|n \\ d<n}} \Phi_d(z)$. By the induction hypothesis, the polynomial $Q(z)$ is in $\mathbb{Z}[z]$. Moreover, since each $\Phi_d(z)$ is monic by definition, their product $Q(z)$ is also monic. Using lemma 2.7 ensures that $\Phi_n(x) \in \mathbb{Q}[z]$ allowing us to conclude: since $z^n - 1 = \Phi_n(z)Q(z)$ where $z^n - 1 \in \mathbb{Z}[z]$ and $Q(z) \in \mathbb{Z}[z]$ is monic, Gauss's Lemma implies that the coefficients of $\Phi_n(z)$ must be integers. ■

Remark 2.14. Finally, a fundamental property of cyclotomic polynomials is their irreducibility over $\mathbb{Q}$. This result guarantee that Φ_n is the minimal polynomial of the primitive n-th roots of unity. While Gauss (1801) [12] was the first to establish this result for prime indices, it was Schönemann (1846) who developed the modern modular approach. As noted by Cox [6], Schönemann discovered the irreducibility criterion, commonly attributed to Eisenstein, while working on the theory of higher congruences and applied it directly to prove this case. The result was later extended to general indices by Kronecker (1854) [15] and Dedekind (1857) [7]. We refer the interested reader to [13] for a complete demonstration.

The next proposition synthetizes some elementary identities of the n-th cyclotomics polynomials. They can be proven either by using corollary 2.11 or checking that both sides have the same zeroes, see [19] and [29].

Proposition 2.15. *For every positive integer n and every prime number p, we have that:*

(i) $\Phi_{pn}(z) = \Phi_n(z^p)$ *if* $p \mid n$

(ii) $\Phi_{pn}(z) = \Phi_n(z^p)/\Phi_n(z)$ *if* $p \nmid n$

(iii) $\Phi_{2n}(z) = (-1)^{\varphi(n)}\Phi_n(-z)$ *if* $2 \nmid n$

(iv) $\Phi_n(z) = \Phi_{\mathrm{rad}(n)}(z^{n/\mathrm{rad}(n)})$

(v) $\Phi_n(1/z) = z^{-\varphi(n)}\Phi_n(z)$ *if* $n > 1$

Remark 2.16. Firstly note that *(v)* showcases that n-th cyclotomics are reciprocal (with the constant being -1 for $n = 1$ and $+1$ for $n > 1$). Secondly, using *(i)* and *(ii)* one can start from $\Phi_1(z) = z - 1$ to inductively compute the n-th cyclotomic polynomials.

2.2 Cyclotomic Polynomials

We are now ready to introduce the general class of cyclotomic polynomials. It is worth noting that in the literature, the terms *cyclotomic polynomial* and *n-th cyclotomic polynomial* are sometimes used interchangeably. This will not be the case in this text, as we follow the convention of McKee and Smyth [14].

Definition 2.17. A *cyclotomic* polynomial is a monic integer polynomial whose zeros are all roots of unity.

Proposition 2.18. *Cyclotomic polynomials are products of polynomials Φ_n for a multiset of values of n.*

Proof. Let $P(z) \in \mathbb{Z}[z]$ be a monic polynomial whose roots are roots of unity. Let ω be a root of P. Then ω is a primitive n-th root of unity for some n. Since $\Phi_n(z)$ is the minimal polynomial of ω over $\mathbb{Q}$ (remark 2.14), it follows that $\Phi_n(z)$ divides $P(z)$. Writing $P(z) = \Phi_n(z)Q(z)$, we observe that $Q(z)$ remains a monic integer polynomial with roots of unity. By induction on the degree, $P(z)$ factors uniquely as a product of cyclotomic polynomials. ∎

2.3 Behavior of Coefficients

Denote the n-th cyclotomic as $\Phi_n(z) = \sum a_n(j)z^j$, so that $a_n(j)$ is the coefficient of z^j in $\Phi_n(z)$. Below are the explicit forms of the first seven n-th cyclotomic polynomials:

$$\begin{aligned}
\Phi_1(z) &= z - 1 \\
\Phi_2(z) &= z + 1 \\
\Phi_3(z) &= z^2 + z + 1 \\
\Phi_4(z) &= z^2 + 1 \\
\Phi_5(z) &= z^4 + z^3 + z^2 + z + 1 \\
\Phi_6(z) &= z^2 - z + 1 \\
\Phi_7(z) &= z^6 + z^5 + z^4 + z^3 + z^2 + z + 1
\end{aligned}$$

Beyond the specific values of the coefficients, the cyclotomic polynomials possess a structural symmetry. Since $\Phi_n(1/z)z^{\varphi(n)} = \Phi_n(z)$ for $n > 1$ (see Proposition 2.8 (v)), the polynomial is palindromic. Consequently, the coefficients satisfy the relation:

$$a_n(\varphi(n) - j) = a_n(j) \quad \text{for } j = 0, \ldots, \varphi(n).$$

This symmetry halves the complexity of computing or analyzing the coefficients. To understand the transition from simple coefficients to larger values, it is useful to classify n by the number of its distinct odd prime factors. We say Φ_n is *binary* if $n = pq$ with $p < q$ primes. Migotti (1883) [23] and Bang (1895) [2] proved that for any binary cyclotomic polynomial, the coefficients lie in $\{-1, 0, 1\}$. A complete description of these coefficients is given by Lam and Leung (1996) [16].

The first counterexample to this smallness occurs at $n = 105$, where -2 appears as a coefficient:

$$\begin{aligned}
\Phi_{105}(z) = {} & z^{48} + z^{47} + z^{46} - z^{43} - z^{42} - 2z^{41} - z^{40} - z^{39} \\
& + z^{36} + z^{35} + z^{34} + z^{33} + z^{32} + z^{31} - z^{28} - z^{26} - z^{24} - z^{22} - z^{20} \\
& + z^{17} + z^{16} + z^{15} + z^{14} + z^{13} + z^{12} - z^{9} - z^{8} - 2z^{7} - z^{6} - z^{5} \\
& + z^2 + z + 1
\end{aligned}$$

The case $n = 105 = 3 \cdot 5 \cdot 7$ represents the first *ternary* cyclotomic polynomial (product of three distinct odd primes). While coefficients here exceed 1 in absolute value, they still exhibit a constrained behavior. Gallot and Moree proved that for any ternary integer n, neighboring coefficients differ by at most one (the *jump one property*) [25]:

$$|a_n(j) - a_n(j-1)| \leq 1.$$

This implies that the set of coefficients for a ternary polynomial is always a set of consecutive integers. However, specific results for cyclotomic polynomials of order greater than three are scarce, and this regularity is not guaranteed to hold for higher orders [25].

To conclude this introduction to the n-th cyclotomics polynomials, let's appreciate the cancellation taking place in the coefficients by quoting D. H. Lehmer [18]:

> The smallness of $|a_n(j)|$ would appear to be one of the fundamental conspiracies of the primitive n-th roots of unity. When one considers that $a_n(j)$ is a sum of $\binom{\varphi(n)}{j}$ unit vectors (for example 73′629′072 in the case of $n = 105$, $j = 7$) one realizes the extent of the cancellation that takes place.

However, despite the significant cancellation described by Lehmer, the coefficients remain unbounded. In fact, Suzuki proved that for any integer $m \in \mathbb{Z}$, there exist n, k such that the coefficient $a_n(k)$ is exactly m [27]. The precise growth rate was established by Bateman and Vaughan [25], let's denote $A(n) = \max_j |a_n(j)|$.

Theorem 2.19 (Bateman-Vaughan)**.** *On the one hand, for every $\varepsilon > 0$, we have*

$$A(n) < \exp\left(n^{(\log 2+\varepsilon)/\log\log n}\right)$$

for all sufficiently large positive integers n. On the other hand, we have

$$A(n) > \exp\left(n^{\log 2/\log\log n}\right)$$

for infinitely many positive integers n.

This theorem reveals a contrast: while roots of unity are geometrically simple, their minimal polynomials can have massive coefficients. This suggests that the standard form $\Phi_n(z)$ is not the most efficient representation in terms of size. This motivates the study of *length* and *cyclotomic shortness*: can we minimize the length by multiplying by cyclotomic factors? The answer is yes. As showcased by Example 3.3, a direct application of Lemma 2.10, every n-th cyclotomic polynomial has a shortness of exactly 2. This is further discussed in the next section.

3 Length and Shortness

In this section, we introduce the central metric of this thesis: the *length* of a polynomial. While the coefficients of cyclotomic polynomials have been studied for their individual growth, the length offers a global perspective on the "size" of the polynomial. We will define the notion of *cyclotomic shortness* and discuss its relationship with other measures of complexity in algebraic number theory, specifically the *Mahler measure* and the *house* of an algebraic integer. Finally, we will establish a reduction result that justifies our focus on the specific family Φ_1^N. We begin by formally defining the length and the shortness of a polynomial.

Definition 3.1. Let $P(z) = \sum_{k=0}^{d} a_k z^k \in \mathbb{C}[z]$. The *length* of P, denoted by $L(P)$, is the sum of the absolute values of its coefficients:

$$L(P) = \sum_{k=0}^{d} |a_k|$$

This quantity is sometimes referred to as the L_1-norm of the coefficient vector. We are often interested in how much this length can be reduced by multiplying by a suitable polynomial. If the polynomial in question is cyclotomic, then the notion of *shortness* appears:

Definition 3.2. Let $P(z) \in \mathbb{Z}[z]$. The *shortness* (or *cyclotomic shortness*) of P, denoted by $\mathrm{sh}(P)$, is the minimum length achievable by multiplying P by a cyclotomic polynomial. Formally:

$$\mathrm{sh}(P) = \inf \{L(PQ) : Q \text{ is cyclotomic}\}$$

A cyclotomic polynomial Q that achieves this minimum, i.e. $L(PQ) = \mathrm{sh}(P)$, is called a *short polynomial* for P.

Example 3.3. Any n-th cyclotomics has length 2. This result is an immediate consequence of Lemma 2.10 and the fact that the shortness of a non zero polynomial must be at least 2. Indeed, as the resulting polynomial retains the roots of the original, its length cannot be 1.

The study of polynomial length is intimately related to the geometry of roots. To understand the utility of shortness, we must introduce two key quantities often arising in this context: the Mahler measure and the house.

Definition 3.4. Let $P(z) = a_d \prod_{j=1}^{d}(z - \alpha_j) \in \mathbb{C}[z]$ with $a_d \neq 0$. The *Mahler measure* of P, denoted $M(P)$, is defined as:

$$M(P) = |a_d| \prod_{j=1}^{d} \max(1, |\alpha_j|)$$

Restricting ourselves to polynomials in $\mathbb{Z}[z]$, a central open problem arose from Lehmer's work in 1933 [17]. He was interested in finding the smallest possible Mahler measure greater than 1. The best candidate he found was the polynomial:

$$L(z) = z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1$$

The measure of this polynomial is $M(L) \approx 1.17628$, a value now known as *Lehmer's number*. Lehmer asked whether there exists a constant $c > 1$ such that $M(P) \geq c$ for all integer polynomials having $M(P) > 1$. The hypothesis that such a gap exists is widely believed to be true and is known as *Lehmer's Conjecture*. In the case where the Mahler measure is below 1.3, it is said to be *small*. While the Mahler measure considers the product of roots outside the unit disk, the *house* considers the maximum modulus.

Definition 3.5. The *house* of an algebraic integer α is the maximum modulus of its conjugates, denoted by $|\overline{\alpha}|$. More generally, the house of a polynomial P, denoted $\left|\overline{P}\right|$, is the maximum modulus of its roots:

$$\left|\overline{P}\right| = \max_{1 \leq j \leq d} |\alpha_j|$$

Remark 3.6. Note that the house of an algebraic number is equal to the house of its minimal polynomial. Furthermore, by Kronecker's Theorem (Theorem 1.3 and 1.4 of [14]), if an algebraic integer is not a root of unity, then its house is greater than 1.

The connection between these concepts and shortness is crucial. Since the roots of any cyclotomic polynomial Q lie on the unit circle, we have $M(Q) = 1$. Consequently, the Mahler measure is preserved under multiplication: $M(PQ) = M(P)$. As discussed by El-Serafy and McKee [9], finding a multiple with minimal length allows us to obtain a more compact representation of an algebraic number without altering its Mahler measure. In essence, shortness serves as a way to reduce the complexity of a polynomial while preserving its geometric size. Notably, for all known small Mahler measures, the minimal polynomial has some multiple with length at most 14, and in the majority of cases, a length of only 5 [9].

It is worth noting that some authors, such as El-Serafy and McKee [9], also define a *reciprocal shortness*, where the multiplier Q is only required to be a monic reciprocal polynomial rather than strictly cyclotomic. This distinction is algorithmically significant. Directly computing cyclotomic shortness is often impractical; however, algorithms exist (e.g., adapted from Filaseta et al. [10]) to efficiently search for reciprocal multiples. Since any cyclotomic polynomial ($n > 1$) is reciprocal, reciprocal shortness serves as a lower bound for cyclotomic shortness. In practice, this approach is highly effective because if the shortest reciprocal multiplier found turns out to be cyclotomic, then the cyclotomic shortness has been determined, which is guaranteed for very short lengths (specifically length ≤ 4, see Remark 2.1 in [9]).

Remark 3.7. Note that for cyclotomic polynomials, all roots lie on the unit circle ($|\alpha_j| = 1$), implying that $M(\Phi_n) = 1$ and $\left|\overline{\Phi_n}\right| = 1$.

3.1 Computability and the House Barrier

The strategy to compute the shortness $sh(P)$ depends drastically on the location of the roots of P. When the polynomial P has a house bounded away from 1, the problem of computing the shortness is well-behaved. A result by Dobrowolski [8] implies that the shortness is effectively computable. More recently, El-Serafy and McKee proved that for any $\epsilon > 0$, the number of monic reciprocal polynomials with a fixed shortness and $\left|\overline{P}\right| \geq 1 + \epsilon$ is finite [9]. This allows for algorithmic search strategies to classify polynomials with small shortness in this regime.

However for all cyclotomic polynomials, including our main subject $\Phi_1^N(z) = (z-1)^N$, the house is always 1. The finiteness results mentioned above collapse, and the standard algorithms used to bound the degree of the short polynomial Q fail. Consequently, we cannot easily bound the search space for Q. This places the determination of $\mathrm{sh}((z-1)^N)$ in a wild territory where the majority of the literature does not bring practicable results.

3.2 Analytic Characterization

We begin with a simple but useful result about the length of a polynomial:

Proposition 3.8. *Set $P \in \mathbb{Z}[z]$, one can bound its length, as follow:*

$$L(P) \geq \sup_{z \in S_1} |P(z)|$$

Proof. Using the classical triangular inequality, we have for all $z \in S_1$:

$$|P(z)| = |\sum_k c_k z^k| \leq \sum_k |c_k||z^k| = \sum_k |c_k| = L(P)$$

■

Corollary 3.9. *Set $Q \in \mathbb{Z}[z]$ a short polynomial for $P \in \mathbb{Z}[z]$, then:*

$$\mathrm{sh}(P) \geq \sup_{z \in S_1} |Q(z)||P(z)|$$

Proof. Direct application of Proposition 3.8 and using the fact that $L(PQ) = \mathrm{sh}(P)$. ■

This result can give us some insight into the structure of the short polynomial Q. E.g. if one has an upper bound for $\mathrm{sh}(P)$, the behaviour of Q and P needs to remain in a certain range in some way. This intuition is what lies behind the result of Proposition 4.8 in Section 4.2.

The definition of the Mahler measure given above requires to know the roots of the polynomial. However, a fundamental result due to Jensen provides an equivalent analytic form, expressing the measure as an integral over the unit circle.

Proposition 3.10. *Let $P(z) \in \mathbb{C}[z]$ be a non-zero polynomial. Then:*

$$M(P) = \exp\left(\int_0^1 \log\left|P(e^{2\pi i t})\right| \, dt\right)$$

In other words, the Mahler measure is the geometric mean of $|P(z)|$ on the unit circle.

Proof. See [14] (or standard texts on analytic number theory). The equality follows from the fact that for any $\alpha \in \mathbb{C}$, $\int_0^1 \log|e^{2\pi i t} - \alpha| \, dt = \log(\max(1, |\alpha|))$. Summing these terms for the roots of P yields the result. ■

This characterization highlights why the Mahler measure is a natural measure of "size" for polynomials. It allows us to relate $M(P)$ directly to the length $L(P)$.

Proposition 3.11. *Let $P(z) \in \mathbb{C}[z]$ be a polynomial of degree d. The length of P and its Mahler measure satisfy the following inequality:*

$$M(P) \leq L(P) \leq 2^d M(P)$$

Proof. From Proposition 3.10, $M(P)$ is the geometric mean of $|P(z)|$ on the unit circle. Since the geometric mean is always bounded by the maximum, we have:

$$M(P) \leq \sup_{|z|=1} |P(z)|$$

Using Proposition 3.8 and combining with the result above yield:

$$M(P) \leq L(P)$$

Now for the upper bound. Let $P(z) = a_d \prod_{j=1}^{d}(z - \alpha_j)$. By Vieta's formulas, the coefficient a_{d-k} is given by:

$$a_{d-k} = (-1)^k a_d \sum_{1 \leq j_1 < \cdots < j_k \leq d} \alpha_{j_1} \ldots \alpha_{j_k}.$$

Taking absolute values and bounding the product of the roots by $\frac{M(P)}{|a_d|}$ yields:

$$|a_{d-k}| \leq |a_d| \sum_{1 \leq j_1 < \cdots < j_k \leq d} \prod_{m=1}^{k} |\alpha_{j_m}| \leq |a_d| \binom{d}{k} \frac{M(P)}{|a_d|} = \binom{d}{k} M(P)$$

Summing over all coefficients:

$$L(P) = \sum_{k=0}^{d} |a_k| \leq M(P) \sum_{k=0}^{d} \binom{d}{k} = 2^d M(P)$$

■

3.3 Reduction to Φ_1^N

Given the difficulty of the problem when $\left|\overline{P}\right| = 1$, one might wonder why we focus specifically on powers of Φ_1. The answer lies in a powerful reduction result proposed as an exercise in [14]. It states that $(z-1)^N$ acts as a worst-case scenario for cyclotomic shortness.

Proposition 3.12. *Set P a cyclotomic polynomial and let N be the largest multiplicity of any zero of P. Then we have:*

$$\mathrm{sh}(P) \leq \mathrm{sh}(\Phi_1^N)$$

Proof. Denote the cyclotomic polynomial as $P = \prod_{j=1}^{J} \Phi_{n_j}^{d_j}$. Note that d_j indicates the multiplicities of the zeros of Φ_{n_i} and thus $N = \max_j d_j$. Let's consider $M = \mathrm{lcm}\left(\{n_j\}_{j=1}^{J}\right)$ the least common multiple of the n_j's and note that by lemma 2.10, we have that:

$$z^M - 1 = \prod_{d|M} \Phi_d(z)$$

By the definition of M, all n_j's divides M and thus appear in this product. This showcases that:

$$\prod_{j=1}^{J} \Phi_{n_j}^{d_j} \mid \prod_{j=1}^{J} \Phi_{n_j}^{N} \mid (z^M - 1)^N$$

showcasing that there exist a cyclotomic Q such that $PQ(z) = (z^M - 1)^N$, giving us the inequality:

$$\mathrm{sh}(P) \leq L(PQ) \leq \mathrm{sh}\left((z^M - 1)^N\right)$$

The last step consist of showing that the shortness of $(z^M - 1)^N$ is smaller or equal to the shortness of Φ_1^N. Set T a short polynomial for Φ_1^N and define the polynomial $\tilde{T}(z) = T(z^M)$ and similarly define $\tilde{\Phi}_1(z) = \Phi_1(z^M)$. Note that $\tilde{T}$ is a monic polynomial having all its roots on the unit circle and is thus a cyclotomic polynomial. Also note that the length of a polynomial does not change under the transformation $z \mapsto z^M$ and thus: $L(T\phi_1^N) = L(\tilde{T}\tilde{\Phi}_1^N)$. This concludes the proof as:

$$\mathrm{sh}\left((z^M - 1)^N\right) \leq L(\tilde{T}\tilde{\Phi}_1^N) = L(T\phi_1^N) = \mathrm{sh}(\Phi_1^N)$$

■

Remark 3.13. This result is rather peculiar. Consider the case where $P = \Phi_1^N$. Multiplying by any cyclotomic polynomial with roots of multiplicities less than N would not change the shortness. In other words it is possible to find short polynomials for Φ_1^N with any n-th cyclotomic Φ_n possible as long as we do not overcharge the multiplicities. As a consequence, for any integer D, it is possible to find a short polynomial of degree greater than D. Note that the short polynomial for Φ_1^N is thus not unique, and in fact there is an infinite number of such short polynomial.

This proposition is fundamental to our work. It implies that if we can understand the shortness of Φ_1^N, we implicitly bound the shortness of all cyclotomic polynomials with root multiplicity bounded by N. the exact answer is unknown and is formally stated as an open problem by McKee and Smyth:

Problem 3.14 (Based on Problem 1.38 of [14])**.** Determine $\mathrm{sh}((z-1)^N)$ for $N \in \mathbb{N}^*$.

4 Shortness of Φ_1^N

We now focus on the central object of this thesis: the polynomial $\Phi_1^N(z) = (z-1)^N$. As established in Proposition 3.12, determining the behavior of this specific family is key to bounding the shortness of any cyclotomic polynomial. Note that the naive length $L((z-1)^N)$ grows exponentially. Indeed by the Binomial Theorem, we have the expansion:

$$\Phi_1^N(z) = (z-1)^N = \sum_{k=0}^{N} \binom{N}{k} (-1)^{N-k} z^k$$

Since the length $L(P)$ is the sum of the absolute values of the coefficients, we obtain:

$$L\left(\Phi_1^N\right) = \sum_{k=0}^{N} \left| \binom{N}{k} (-1)^{N-k} \right| = \sum_{k=0}^{N} \binom{N}{k} = 2^N$$

We rely on empirical evidence to build intuition. In this section, we present values obtained through computational search algorithms. These results illustrate the growth of the shortness and motivate the theoretical bounds derived later.

Since one cannot impose a global bound on the degree of the short polynomial, a direct computation of the shortness is not feasible at this point. Instead, we approximate the result by searching for a cyclotomic multiplier Q of a fixed degree D. We know that Q is of the form $\prod_{n \in \mathcal{N}} \Phi_n$, where $\mathcal{N}$ is a multiset of integers. Since $\deg(\Phi_n) = \varphi(n)$, fixing the degree of Q to D is equivalent to finding all multisets $\mathcal{N}$ satisfying:

$$\sum_{n \in \mathcal{N}} \varphi(n) = D$$

We perform this search for $D \in \{1, \ldots, D_{max}\}$. To limit the search space for a given D, we only consider integers n such that $\varphi(n) \leq D$. Using the lower bound $\varphi(n) \geq \sqrt{n/2}$, we can restrict our search to $n \leq 2D^2$.

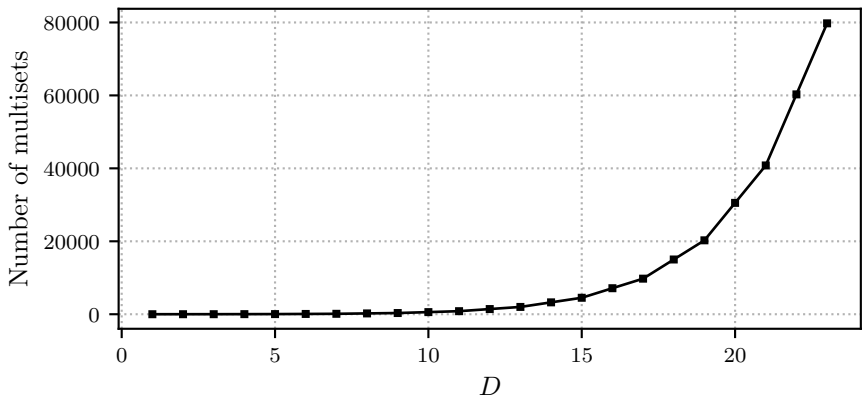

Figure 1: Evolution of the number of multisets in function of the degree D. In other words the y-axis represent the number of existing cyclotomic polynomials of degree D.

This combinatorial task was performed on the Helvetios cluster. The problem is computationally heavy and resembles the integer partition problem. Since the totient function is not surjective, standard partition bounds do not apply directly. However, one can establish a lower bound on the number of multisets by restricting the admissible parts to powers of two. Since $\varphi(2^{k+1}) = 2^k$ for all $k \geq 0$, every power of two is a realizable totient value. By ignoring other totient values, the problem reduces to enumerating the binary partitions of D, denoted by $b(D)$. The total count is therefore bounded below by $b(D)$, which satisfies the asymptotic growth $b(D) \sim D^{\frac{\ln D}{2\ln 2}}$ [20].

Although this lower bound is loose (e.g., predicting at least 647 multisets for $D = 20$ while the true value is $30'532$), it shows that the search space grows rapidly. The actual number of multisets grows much faster, as shown in Figure 1, see Figure 5 in the appendix A for an exponential fit. The growth is approximately exponential with a growth rate approximately 0.46. Due to these constraints, we stopped the search at $D_{max} = 23$ with $79'762$ multisets, as the computation for higher degrees exceeded the time limits available on the cluster.

This provides the number of available cyclotomic polynomials for each degree D, as well as the multisets defining them. The strategy is to compute the length of the product $L(\Phi_1^N \prod_{n\in\mathcal{N}} \Phi_n)$ for each degree and each multiset $\mathcal{N}$. The resulting lengths are compared, and only the multisets yielding the minimal length are retained. A detailed summary of these results is provided in Appendix B. Since the exact shortness is unknown, this method yields a directly computable upper bound for $\mathrm{sh}(\Phi_1^N)$. Figure 2 displays the minimal lengths found for $N \in \{1, \ldots, 30\}$ on a logarithmic scale, alongside the trivial bound 2^N. As shown, the computed bound is significantly better than the trivial one.

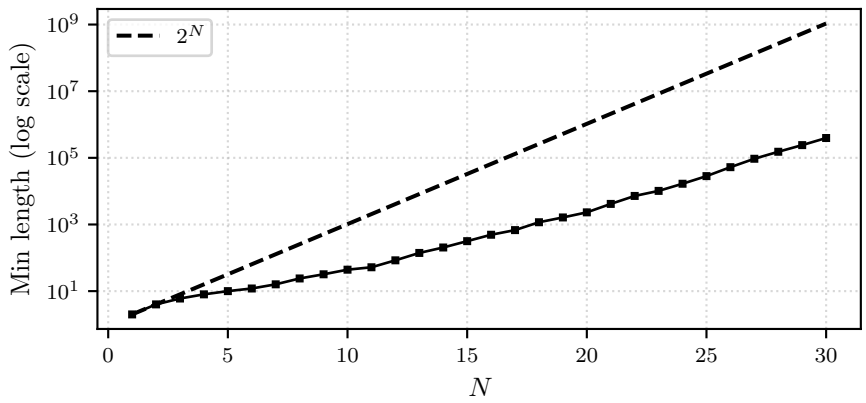

Figure 2: Minimal length of the product $L(\Phi_1^N \prod_{n\in\mathcal{N}} \Phi_n)$ found for cyclotomic polynomials of degree up to 23, as a function of N. The y-axis is logarithmic, and the trivial bound 2^N is plotted for comparison.

For the first three values $N \in \{1, 2, 3\}$, this bound matches the actual shortness, as verified in Section 4.2 where these values are explicitly computed. For $N > 3$, it is impossible to determine if the true shortness is reached. However, numerical evidence

suggests that the bound remains valid for $N \leq 6$. The minimal length consistently equals $2N$ in this range, and since the shortness is constrained to be even (Proposition 4.5), descending below this threshold appears to be difficult. For a clearer visualization of the initial growth ($N \leq 10$), we refer to Figure 6 in the appendix.

An analysis of the optimal multisets $\mathcal{N}$ reveals a notable structural pattern related to divisibility. Heuristically, significant length reduction often occurs when cyclotomic factors combine to form sparse polynomials like $z^m - 1$. As established by Lemma 2.10, the product of cyclotomic polynomials over all divisors of m satisfies $\prod_{d|m} \Phi_d(z) = z^m - 1$, which has a minimal length of 2. Consequently, a natural strategy to minimize the length is to ensure that for any index m included in the multiset, all its divisors are also present. We refer to a multiset satisfying this condition as "closed". Our computational results strongly support this heuristic: among all the best multisets identified, only a single counter-example violates this rule (at $N = 4$, where the index 12 appears without 6).

4.1 Theoretical Bounds

While the computational results presented in the previous section provide exact values for small N and illustrate the initial growth of the shortness, they are limited by the exponential expansion of the search space. In this section, we establish rigorous bounds on the shortness of Φ_1^N. We begin by addressing the upper bound. As noted earlier, the trivial expansion of $(z-1)^N$ yields a length of 2^N. However, by explicitly constructing a multiple using the second cyclotomic polynomial Φ_2, we demonstrate that the shortness grows slower than the naive bound.

Proposition 4.1. *For all $N \in \mathbb{N}^*$ we have the upper bound:*

$$\mathrm{sh}(\Phi_1^N) \leq 2 \binom{N}{\lfloor N/2 \rfloor}$$

Proof. We want to bound the length of $\Phi_1^N \Phi_2$. Using $\Phi_2(z) = z + 1$ and expanding the binomial $(z-1)^N$:

$$\begin{aligned}
\Phi_1^N(z)\Phi_2(z) &= (z-1)^N(z+1) \\
&= (z+1)\sum_{k=0}^{N}\binom{N}{k}(-1)^{N-k}z^k \\
&= \sum_{k=0}^{N}\binom{N}{k}(-1)^{N-k}z^{k+1} + \sum_{k=0}^{N}\binom{N}{k}(-1)^{N-k}z^k \\
&= \sum_{k=1}^{N+1}\binom{N}{k-1}(-1)^{N-k+1}z^k + \sum_{k=0}^{N}\binom{N}{k}(-1)^{N-k}z^k \\
&= \sum_{k=0}^{N+1}\left(\binom{N}{k} - \binom{N}{k-1}\right)(-1)^{N-k}z^k
\end{aligned}$$

Using the convention $\binom{N}{-1} = \binom{N}{N+1} = 0$. One can thus compute the length as follow:

$$L(\Phi_1^N\Phi_2) = \sum_{k=0}^{N+1}\left|\binom{N}{k} - \binom{N}{k-1}\right|$$

The key observation here is that the binomial $\binom{N}{k}$ is an increasing function until $k = \lfloor N/2 \rfloor$ and then becomes a decreasing function. Thus one can get rid of the absolute value of the sum and use the telescoping cancellation to explicitly compute:

$$\begin{aligned} L(\Phi_1^N \Phi_2) &= \sum_{k=0}^{\lfloor N/2 \rfloor} \left[\binom{N}{k} - \binom{N}{k-1} \right] + \sum_{k=\lfloor N/2 \rfloor + 1}^{N+1} \left[\binom{N}{k-1} - \binom{N}{k} \right] \\ &= -\binom{N}{-1} + \binom{N}{\lfloor N/2 \rfloor} + \binom{N}{\lfloor N/2 \rfloor} - \binom{N}{N+1} \\ &= 2\binom{N}{\lfloor N/2 \rfloor} \end{aligned}$$

■

Corollary 4.2. *We have:*

$$\mathrm{sh}(\Phi_1^N) = O\left(\frac{2^N}{\sqrt{N}} \right)$$

Proof. To analyze the asymptotic behavior of this bound as $N \to \infty$, we rely on Stirling's approximation for the central binomial coefficient:

$$\binom{N}{\lfloor N/2 \rfloor} \sim \frac{2^N}{\sqrt{\pi N/2}}$$

Substituting this into our result yields the following asymptotic estimate:

$$\mathrm{sh}(\Phi_1^N) \lesssim \sqrt{\frac{8}{\pi}} \frac{2^N}{\sqrt{N}}$$

■

In terms of order of magnitude, this establishes that the shortness grows slower than the trivial bound 2^N. Conversely, we also derive a lower bound. Since any short polynomial implies a multiple with a root at $z = 1$ of high multiplicity, constraints on the number of non-zero coefficients allow us to establish a minimum growth rate. Initially, this result was established using heavy computations involving the linear independence of column of Vandermonde-like matrices. However, Paul Erdős would surely argue that such a proof is not "from the Book." Consequently, we present here a much simpler and more elegant approach based on [5, 21]:

Proposition 4.3. *Let α be a root of multiplicity N for a polynomial $P \in \mathbb{C}[z]$. Then it has at least $N + 1$ nonzero coefficients.*

Proof. Let's denote the polynomial as $P = c_n z^{k_n} + \cdots + c_1 z_1^k$ and let's consider a slightly modified version $\tilde{P} = c_n z^{k_n - k_1} + \cdots + c_1$. Note that the weight of the polynomial is unchanged, i.e. $w(\tilde{P}) = w(P)$. Of course $(z - \alpha)^N | P(z)$ for some $\alpha \neq 0$ entails $(z - \alpha)^N | \tilde{P}(z)$. By formal differentiation:

$$(z - \alpha)^N \;\Big|\; c_n z^{k_n - k_1} + \cdots + c_2 z^{k_2 - k_1} + c_1$$

implies

$$(z - \alpha)^{N-1} \;\Big|\; c_n (k_n - k_1) z^{k_n - k_2} + \cdots + c_2 (k_2 - k_1)$$

and the claim follows by induction on N. ■

Corollary 4.4. *For all $N \in \mathbb{N}^*$ we have the lower bound:*

$$\mathrm{sh}(\Phi_1^N) \geq N + 1$$

Proof. Proposition 4.3 immediately gives us a lower bound on the shortness of Φ_1^N as the weight is smaller or equal to the length. Setting Q a short polynomial, gives us:

$$\mathrm{sh}(\Phi_1^N) = L(\Phi_1^N Q) \geq w(\Phi_1^N Q) \geq N + 1$$

■

Figure 7 in Appendix A compares the experimental values found by our algorithm with the theoretical bounds established in this section. It appears that the theoretical bounds are loose, particularly the upper bound. The tightness of the lower bound is harder to assess, as the computational results provide only an upper bound on the true shortness (which is not necessarily achieved for $N > 3$).

4.2 Miscellaneous Results

While establishing asymptotic bounds is crucial, examining small values of N provides concrete insight into how cyclotomic factors interact with $(z-1)^N$. In this section, we explicitly derive the shortness for $N = 1, 2, 3$. Note that for $N = 1$ Example 3.3 already established that the shortness is 2. We start with a small useful result.

Proposition 4.5. $\mathrm{sh}(\Phi_1^N)$ *is an even number.*

Proof. This proof relies on the fact that a number $x \in \mathbb{Z}$ and its modulus $|x|$ shares the same parity, i.e. they have the same projection in $\mathbb{Z}_2$. Set Q a short poly for Φ_1^N and denote $\Phi_1^N Q(z) = \sum_k c_k z^k$. As 1 is a root of such polynomial, we have:

$$0 = \Phi_1^N Q(1) = \sum_k c_k \equiv \sum_k |c_k| \pmod 2$$

note that the last term is the length of $\Phi_1^N Q$ and thus the shortness of Φ_1^N. ■

Proposition 4.6. *For $N = 2$ we have:*

$$\mathrm{sh}(\Phi_1^2) = 4$$

Proof. Firstly using the lower bound (Corollary 4.4) one can establish that $\mathrm{sh}(\Phi_1^2) \geq 3$ but as the shortness is even by Proposition 4.5 one obtain: $\mathrm{sh}(\Phi_1^2) \geq 4$. However note the computation:

$$\Phi_1^2 \cdot \Phi_2(z) = (z-1)^2(z+1) = z^3 - z^2 - z + 1$$

which is a polynomial of length 4, confirming that Φ_2 is a short polynomial for Φ_1^2. ■

For $N = 3$ the strategy remains essentially the same, but the computation becomes a bit more messy.

Proposition 4.7. *For $N = 3$ we have:*

$$\mathrm{sh}(\Phi_1^3) = 6$$

Proof. First, we establish the upper bound by constructing a suitable cyclotomic multiple. Consider the following cyclotomic product:

$$\begin{aligned} P &= (z^4 - 1)(z^3 - 1)(z - 1) \\ &= z^8 - z^7 - z^5 + z^3 + z - 1 \end{aligned}$$

The length of this polynomial is 6. As $\Phi_1^3 | P$ it showcases that $\mathrm{sh}(\Phi_1^3) \leq 6$. From Corollary 4.4: $\mathrm{sh}(\Phi_1^3) \geq 3 + 1 = 4$. Since the shortness must be an even number (Proposition 4.5), we have $\mathrm{sh}(\Phi_1^3) \in \{4, 6\}$. We now prove that $\mathrm{sh}(\Phi_1^3) \neq 4$ by contradiction.

Assume there exists a polynomial $Q(z) = \sum c_k z^k$ divisible by $\Phi_1(z)^3 = (z - 1)^3$ with length 4. Since Q is divisible by $(z - 1)^3$, it must satisfy $Q(1) = Q'(1) = Q''(1) = 0$, the conditions translate to:

1. $Q(1) = \sum c_k = 0$
2. $Q'(1) = \sum k c_k = 0$
3. $Q''(1) = \sum k(k-1)c_k = \sum k^2 c_k - \sum k c_k = 0 \implies \sum k^2 c_k = 0$

Given that $L(Q) = \sum |c_k| = 4$ and $\sum c_k = 0$, the multiset of non-zero coefficients must be either $\{2, -2\}$, $\{1, 1, -2\}$, or $\{1, 1, -1, -1\}$. We analyze these cases:

(i) *Coefficients* $\{2, -2\}$. $Q(z) = 2z^m - 2z^n$. Condition (2) implies $2m - 2n = 0 \implies m = n$, which means $Q(z) = 0$, a contradiction.

(ii) *Coefficients* $\{1, 1, -2\}$. $Q(z) = z^m + z^n - 2z^p$. Condition (2) gives $m + n - 2p = 0 \implies p = \frac{m+n}{2}$. Condition (3) gives $m^2 + n^2 - 2p^2 = 0$. Substituting p:

$$m^2 + n^2 - 2\left(\frac{m+n}{2}\right)^2 = 0 \implies 2m^2 + 2n^2 - (m^2 + 2mn + n^2) = 0$$

$$\implies (m - n)^2 = 0 \implies m = n$$

This implies terms combine, reducing the length, which is not admissible.

(iii) *Coefficients* $\{1, 1, -1, -1\}$. The polynomial takes the form $Q(z) = z^m + z^n - z^p - z^q$. The vanishing of the first and second derivatives at $z = 1$ imposes the following system on the exponents:

$$\begin{cases} m + n = p + q \\ m^2 + n^2 = p^2 + q^2 \end{cases}$$

Consider the auxiliary quadratic polynomials $A(X) = (X - m)(X - n)$ and $B(X) = (X - p)(X - q)$. Expanding them yields:

$$A(X) = X^2 - (m + n)X + mn \quad \text{and} \quad B(X) = X^2 - (p + q)X + pq$$

By the first equation of the system, the coefficients of X are equal: $m + n = p + q$. For the constant terms, we use the identity $2xy = (x + y)^2 - (x^2 + y^2)$. We deduce:

$$2mn = (m + n)^2 - (m^2 + n^2) = (p + q)^2 - (p^2 + q^2) = 2pq \implies mn = pq$$

Since $A(X)$ and $B(X)$ have identical coefficients, they are the same polynomial. Consequently, their sets of roots are identical, i.e., $\{m, n\} = \{p, q\}$. Substituting back into $Q(z)$, we find $Q(z) = 0$, a contradiction.

Since all configurations for length 4 lead to contradictions, we conclude $sh(\Phi_1^3) > 4$ and therefore $\mathrm{sh}(\Phi_1^3) = 6$ ■

As discussed in Section 3, Corollary 3.9 yields constraints on the short polynomial. Specifically, it forces certain cyclotomic factors to appear for N large enough. This result relies on the observation that for any $n \neq 2$, the evaluation $|\Phi_n(-1)|$ is a non-zero integer (and thus ≥ 1). This is a direct consequence of cyclotomic polynomials having integer coefficients (Theorem 2.13) and the fact that $\omega_2 = -1$ is not a root of Φ_n for $n \neq 2$.

Proposition 4.8. *For $N \geq 3$, the short polynomial Q for Φ_1^N is divisible by Φ_2.*

Proof. Let $Q(z) = \prod_k \Phi_{n_k}(z)$ be a short polynomial for Φ_1^N. By Corollary 3.9, we have the lower bound:

$$sh(\Phi_1^N) \geq \sup_{z \in S_1} |Q(z)(z-1)^N|$$

Evaluating this inequality at the root of unity $\omega_2 = -1$, and assuming $\Phi_2 \nmid Q$ (so $Q(-1) \neq 0$), we obtain:

$$sh(\Phi_1^N) \geq |Q(-1)| \cdot |-1-1|^N = \left(\prod_k |\Phi_{n_k}(-1)|\right) 2^N$$

Based on the discussion above, since no factor Φ_{n_k} is Φ_2, each term $|\Phi_{n_k}(-1)|$ is a non-zero integer, so their product is at least 1. Thus, we obtain the following bound for the shortness:

$$sh(\Phi_1^N) \geq 2^N$$

This contradicts the asymptotic behavior $O(2^N/\sqrt{N})$ established in Corollary 4.2. Consequently, for sufficiently large N, the short polynomial must eliminate this exponential growth, which implies $\Phi_2 \mid Q$.

Numerically, we observe that this contradiction arises for very small values of N. Comparing the lower bound 2^N against the combinatorial upper bound from Proposition 4.1, the first contradiction appears at $N = 3$: $2^3 = 8 > 6$. Thus, the necessity of the factor Φ_2 is established for all $N \geq 3$. ■

Remark 4.9. Note that the bound $N \geq 3$ is tight. As shown in the table in Appendix B, for $N \in \{1, 2\}$, multisets excluding the index 2 achieve the shortness, whereas Φ_2 is consistently present starting from $N = 3$.

5 Uncertainty principle

Notation Convention. Throughout this section, we will abusively identify an element $g \in \mathbb{Z}_m$ with its canonical representative in $\{0, \dots, m-1\} \subset \mathbb{Z}$.

The main reference for this section is the work of T. Tao [28]. The fundamental motivation behind the uncertainty principle is that a non-zero function cannot be highly concentrated in both physical space and frequency space simultaneously. In the context of our polynomial coefficients, this trade-off provides rigid structural constraints.

Definition 5.1. Let G be a finite abelian additive group. A function $e : G \times G \to S_1$ is called a *non-degenerate bi-character* if it is bilinear and $e(x, \cdot)$ is the trivial character if and only if $x = 0$.

Definition 5.2. For any complex-valued function $f : G \to \mathbb{C}$, we define its Fourier transform $\hat{f} : G \to \mathbb{C}$ by the formula:

$$\hat{f}(\xi) = \frac{1}{|G|} \sum_{x \in G} f(x) \overline{e(x, \xi)}$$

Definition 5.3. The *support* of a function is defined as $\operatorname{supp}(f) = \{x \in G : f(x) \neq 0\}$.

We now state the standard uncertainty principle for finite abelian groups [28].

Theorem 5.4. *For any non-zero function $f : G \to \mathbb{C}$:*

$$|\operatorname{supp}(f)| \cdot |\operatorname{supp}(\hat{f})| \geq |G|$$

Proof. From the definition of the Fourier transform, we have the trivial bound for any ξ:

$$|\hat{f}(\xi)| \leq \frac{1}{|G|} \sum_{x \in G} |f(x)| = \frac{\|f\|_1}{|G|}$$

Using the fact that the sum has only $|\operatorname{supp}(f)|$ non-zero terms, we bound the L^1 norm by the L^∞ norm:

$$\|f\|_1 \leq |\operatorname{supp}(f)| \sup_x |f(x)|$$

Combining these gives $\sup_\xi |\hat{f}(\xi)| \leq \frac{|\operatorname{supp}(f)|}{|G|} \sup_x |f(x)|$. Similarly, by the Fourier inversion formula $f(x) = \sum \hat{f}(\xi) e(x, \xi)$, we have:

$$\sup_x |f(x)| \leq \|\hat{f}\|_1 \leq |\operatorname{supp}(\hat{f})| \sup_\xi |\hat{f}(\xi)|$$

Substituting the first inequality into the second yields:

$$\sup_x |f(x)| \leq |\operatorname{supp}(\hat{f})| \frac{|\operatorname{supp}(f)|}{|G|} \sup_x |f(x)|$$

Since f is non-zero, we can divide by $\sup |f(x)|$, obtaining $|G| \leq |\operatorname{supp}(f)| \cdot |\operatorname{supp}(\hat{f})|$. ∎

Proposition 5.5. *Set Q a short polynomial for Φ_1^N. Set $m \in \mathbb{N}$ such that Φ_m is in Q but $(z^m - 1) \nmid Q\Phi_1^N$. Then we have the inequality:*

$$\operatorname{sh}(\Phi_1^N) \geq \frac{m}{m - \varphi(m)}$$

Proof. The strategy is to use the uncertainty principle to obtain an inequality on m. Let $Q\Phi_1^N(z) = \sum_{k\in\mathbb{N}} c_k z^k$. We define a function on $\mathbb{Z}_m$ by wrapping the coefficients:

$$f(g) = \sum_{k\equiv g \pmod m} c_k$$

This corresponds to the coefficients of $Q\Phi_1^N(z) \pmod{z^m - 1}$. Using the standard bi-character $e(x,\xi) = \omega_m^{x\xi}$, the Fourier transform is:

$$\hat{f}(\xi) = \frac{1}{m} Q\Phi_1^N(\omega_m^{-\xi})$$

Since Φ_m divides $Q\Phi_1^N$, the polynomial vanishes at primitive m-th roots of unity. Thus, if $\gcd(\xi, m) = 1$, then $\hat{f}(\xi) = 0$. This bounds the support:

$$|\operatorname{supp}(\hat{f})| \leq m - \varphi(m)$$

For the support of f, we observe that $|f(g)| \geq 1$ if $f(g) \neq 0$ (as coefficients are integers). Thus:

$$|\operatorname{supp}(f)| \leq \sum |f(g)| \leq \sum |c_k| = \operatorname{sh}(\Phi_1^N)$$

Since $(z^m - 1) \nmid Q\Phi_1^N$, f is non-zero. Applying Theorem 5.4:

$$\operatorname{sh}(\Phi_1^N)(m - \varphi(m)) \geq |\operatorname{supp}(f)| \cdot |\operatorname{supp}(\hat{f})| \geq m$$

Dividing by $m - \varphi(m)$ yields the result. ∎

Answering a question posed by Tao, Meshulam established a refined version of the uncertainty principle for finite abelian groups [22]. This result underpins the following corollary. Although no practical application of this bound was found within the scope of this project, we include the result for the sake of completeness.

Corollary 5.6. *Set Q and m exactly as in Proposition 5.5. Set $d_1 \leq |\operatorname{supp}(f)| \leq d_2$ for d_1, d_2 two consecutive divisor of m. Then we have the inequality:*

$$\operatorname{sh}(\Phi_1^N) \geq d_1 d_2 \frac{\varphi(m)}{m} - d_1 d_2 + d_1 + d_2$$

Proof. The proof structure is exactly the same as the Proposition 5.5 but instead of applying the classical uncertainty principle, we apply the improved version by Meshulam, see Theorem [22]. ∎

Corollary 5.7. *Set Q a short polynomial for Φ_1^N and p a prime. If $p > \operatorname{sh}(\Phi_1^N)$ and Φ_{p^k} is present in the cyclotomic Q for some $k \in \mathbb{N}^*$, then the whole chain:*

$$\Phi_p, \Phi_{p^2}, \ldots, \Phi_{p^k}$$

is in the cyclotomic Q as well.

Proof. This corollary is the contrapositive of Proposition 5.5 applied to $m = p^k$. For a prime power, $\frac{m}{m-\varphi(m)} = \frac{p^k}{p^{k-1}} = p$. The condition $p > \operatorname{sh}(\Phi_1^N)$ violates the inequality of Proposition 5.5. Therefore, the premise $(z^m - 1) \nmid Q\Phi_1^N$ must be false. Thus $(z^{p^k} - 1)$ divides $Q\Phi_1^N$. Since $z^{p^k} - 1 = \prod_{j=0}^{k} \Phi_{p^j}$, the entire chain must be present. ∎

Remark 5.8. This result provides intuition regarding the presence of factors of the form $(z^m - 1)$ in short polynomials. The proposition proves this behavior for $m = p^k$ whenever $p > sh(\Phi_1^N)$. Furthermore, as noted in Remark 3.13, it is always possible to construct a short polynomial containing Φ_{p^k}; in this regime, the proposition forces the inclusion of the entire chain, forming the complete divisor $(z^{p^k} - 1)$.

Building on the intuition developed in Remark 5.8 and the numerical patterns observed in Table B, it appears that including factors of the form $z^m - 1$ significantly contributes to minimizing the length. A classical archetype of such behavior is found in the Pentagonal Number Theorem, which relates an infinite product of binomials to a highly sparse series. Motivated by this connection, the following section explores truncated versions of this identity as potential candidates for establishing new bounds on the shortness.

6 The Pentagonal Number Theorem

The study of the shortness of Φ_1^N leads us to consider products of cyclotomic polynomials that produce sparse series. A fundamental archetype of such behavior is found in the work of Euler. He established a surprising identity relating an infinite product of binomials to a sparse power series involving pentagonal numbers [3]. While Euler originally discovered this result experimentally and later proved it algebraically [3], Franklin provided a combinatorial proof in 1881 [11]. This proof is of particular interest to us, as it relies on a cancellation and combinatorial argument that provides insight into the structure of truncated products.

Theorem 6.1 (Euler's Pentagonal Number Theorem)**.** *The following identity holds:*

$$\prod_{k=1}^{\infty}(1-z^k) = \sum_{k=-\infty}^{\infty}(-1)^k z^{\frac{k(3k-1)}{2}} = 1 - z - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots$$

Sketch of Proof. We follow the combinatorial argument by Franklin [11]. The product can be expanded as:

$$\prod_{k=1}^{\infty}(1-z^k) = \sum_{n=0}^{\infty}(p_e(n) - p_o(n))z^n$$

where $p_e(n)$ (resp. $p_o(n)$) denotes the number of partitions of n into an even (resp. odd) number of distinct parts. To determine the coefficients, we establish a bijection between even and odd partitions. We represent a partition with distinct parts using a Ferrers diagram. Let a be the number of dots in the bottom row and b be the number of dots in the rightmost diagonal (see Figure 3). We define a transformation as follows:

If $a \leq b$, we move the bottom row to form a new diagonal.

If $a > b$, we move the diagonal to form a new bottom row.

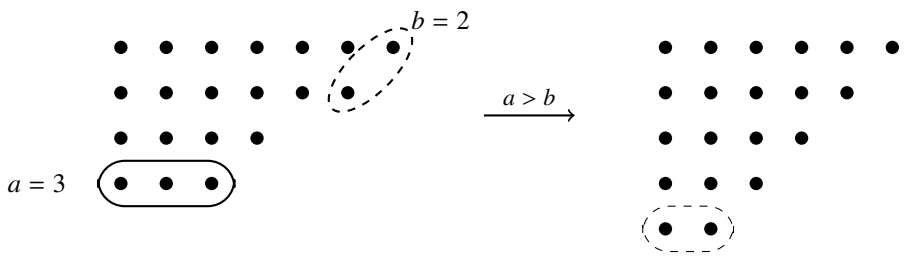

Figure 3: Ferrers diagram illustrating Franklin's involution for the partition $20 = 7 + 6 + 4 + 3$. Here $a = 3$ (bottom row) and $b = 2$ (diagonal). Since $a > b$, the diagonal is moved to form a new row, resulting in the partition $6 + 5 + 4 + 3 + 2$.

This operation changes the number of parts by exactly one, flipping the parity. Thus, for most integers n, the contributions of even and odd partitions cancel out perfectly. The cancellation fails only in the "edge cases" where the intervals overlap ($a = b$ or $a = b + 1$), making the move illegal. These exceptions occur precisely when n is a pentagonal number $n = \frac{k(3k\pm1)}{2}$, leaving a non-zero term $(-1)^k$. ■

Motivated by this theorem, we consider the finite truncation of the pentagonal product.

Definition 6.2. We define the truncated Euler polynomial $Q_N(z)$ as:

$$Q_N(z) = \prod_{k=1}^{N}(1 - z^k)$$

As noted in Section 2, cyclotomic polynomials possess a symmetry in their coefficients. We formalize this property for $Q_N(z)$.

Proposition 6.3. *Let $P(z)$ and $R(z)$ be two polynomials of degree d_P and d_R respectively. If P and R are reciprocal, then their product $P(z)R(z)$ is also reciprocal.*

Proof. By hypothesis, there exist constants c_P, c_R such that $P(1/z) = c_P z^{-d_P} P(z)$ and $R(1/z) = c_R z^{-d_R} R(z)$. Computing the reciprocal of the product:

$$(PR)(1/z) = P(1/z)R(1/z) = c_P c_R z^{-(d_P+d_R)} P(z)R(z)$$

Thus, PR satisfies the definition of reciprocity with constant $c_P c_R$. ■

Corollary 6.4. *The polynomial $Q_N(z)$ is reciprocal of constant $c_{Q_N} = \pm 1$.*

Proof. From Lemma 2.10, we know that $z^k - 1 = \prod_{d|k} \Phi_d(z)$. Thus, we can factor the truncation as a product of cyclotomic polynomials:

$$Q_N(z) = \prod_{k=1}^{N}(1 - z^k) = (-1)^N \prod_{k=1}^{N} \prod_{d|k} \Phi_d(z)$$

From Proposition 2.15, we know that $\Phi_n(z)$ is reciprocal for all n of constant ± 1. Since Q_N is a product of these reciprocal factors, Proposition 6.3 ensures that $Q_N(z)$ is itself reciprocal. ■

6.1 The Failure of Cancellation

To understand the behavior of the coefficients, let us express the truncated product $Q_N(z)$ using the partition notation. The coefficient of z^m can be written as:

$$a_m = \sum_{\substack{\sum_{j=1}^{N} jk_j = m \\ k_j \in \{0,1\}}} (-1)^{\sum_{j=1}^{N} k_j}$$

Here, $k_j = 1$ indicates that the integer j is a part of the partition, and $k_j = 0$ otherwise. The restriction of the truncation corresponds to forcing $k_j = 0$ for all $j > N$.

In the infinite case, Franklin's involution pairs configurations with even $\sum k_j$ to those with odd $\sum k_j$, resulting in complete cancellation (except for pentagonal numbers). In the truncated case, this pairing is only guaranteed if both the original partition and its Franklin image use parts less than or equal to N.

For small degrees $m \leq N$, any partition of m into distinct parts uses only integers smaller than m, so the condition $j \leq N$ is automatically satisfied. The cancellation holds perfectly, and the coefficients a_m match the sparse sequence $0, \pm 1$. However, for $m > N$, a partition may be valid for Q_N (using parts $\leq N$) but its image under Franklin's map might require a part of size $N + 1$. Since this image is excluded from

the sum, the original term remains uncancelled. These "orphan" terms accumulate, causing the coefficients to grow.

The polynomial $Q_N(z)$ has degree $D_N = \frac{N(N+1)}{2}$. Since Q_N is reciprocal, we have the symmetry $|a_m| = |a_{D_N - m}|$. Consequently, the sparsity observed for $m \leq N$ is mirrored at the end of the polynomial. The breakdown of cancellation is thus confined to the middle range:

$$N < m < \frac{N(N+1)}{2} - N$$

In this interval, the coefficients can become very large. We refer to this concentration of mass as the "Sudler Mountain", in reference to Culbreth Sudler Jr. who studied the asymptotic magnitude of these coefficients. Figure 4 showcases this effect for $N = 60$, where some coefficients are bigger than 5000. Figure 8 in Appendix A showcases the same plot but for $N = 500$, with some coefficients reaching a magnitude of 10^{40}.

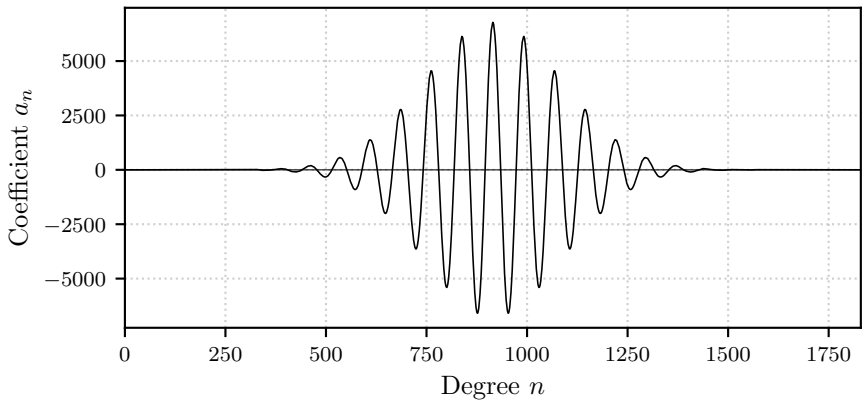

Figure 4: Value of the coefficients of the truncated Euler polynomial Q_{60} in function of the degree n.

It is worth noting that the coefficients do not explode immediately after the threshold N. For the first step $m = N + 1$, the unique forbidden partition is the singleton $\{N + 1\}$. Its contribution to the infinite sum is $(-1)^1 = -1$; removing it simply increases the coefficient by 1. Similarly, for $m = N + 2$, the forbidden partitions are $\{N + 2\}$ and $\{N + 1, 1\}$. Their respective contributions are -1 and $+1$, which cancel out. Thus, the coefficient at $N + 2$ matches the Euler limit exactly. The massive growth characteristic of the Sudler Mountain only begins once the number of forbidden partitions increases drastically, breaking this delicate balance.

6.2 Asymptotic Upper Bound via Sudler's Constant

While the truncation fails to produce perfect cancellation beyond degree N, the resulting polynomial $Q_N(z)$ remains a remarkably efficient candidate for bounding the shortness. Since $Q_N(z)$ is cyclotomic and contains the factor $(1 - z)^N$, its length provides an upper bound:

$$\mathrm{sh}(\Phi_1^N) \leq L(Q_N)$$

To evaluate this length, we examine the growth of the coefficients of $Q_N(z)$. Let M_N denote the maximum absolute coefficient:

$$M_N = \max_m |a_m|$$

The asymptotic behavior of this maximum was studied by Sudler in 1964 [26] and rigorously established by Wright in the same year [30].

Theorem 6.5 (Wright)**.** *There exists a constant K, now known as Sudler's constant, such that:*

$$\log M_N \sim KN \quad \text{as } N \to \infty$$

Numerical approximation yields $K \approx 0.19861$, implying that M_N grows roughly as $(e^K)^N \approx (1.219)^N$.

We can now derive the asymptotic behavior of the length $L(Q_N)$. The length is bounded by the maximum coefficient times the number of terms (the degree plus one):

$$M_N \leq L(Q_N) \leq \left(\frac{N(N+1)}{2} + 1\right) M_N$$

Taking the logarithm and dividing by N:

$$\frac{\log M_N}{N} \leq \frac{\log L(Q_N)}{N} \leq \frac{\log\left(\frac{N(N+1)}{2} + 1\right) + \log M_N}{N}$$

Note that for N sufficiently large we have $\frac{N(N+1)}{2} + 1 \leq N^2$, as such:

$$\frac{\log M_N}{N} \leq \frac{\log L(Q_N)}{N} \leq \frac{\log\left(N^2\right) + \log M_N}{N}$$

As $N \to \infty$, the term $\frac{\log(N^2)}{N}$ vanishes. Using Wright's result, both sides converge to K.

Corollary 6.6. *The shortness of Φ_1^N satisfies the asymptotic upper bound:*

$$\limsup_{N\to\infty} \frac{\log sh(\Phi_1^N)}{N} \leq K \approx 0.19861$$

This result is significant. The trivial upper bound for the shortness is 2^N (growth rate $\log 2 \approx 0.693$). The truncated Euler product provides a construction with growth rate $K \approx 0.19861$, which is exponentially better. The "Sudler Mountain", despite its magnitude, is exponentially smaller than the binomial coefficients of the unreduced polynomial. Note that asymptotically, for the same reasons, it is exponentially better than the bound derived in 4.1 using Φ_2.

7 Final Remarks

This thesis explored the concept of cyclotomic *shortness*, focusing specifically on the polynomial $\Phi_1^N(z) = (z-1)^N$. As established in Proposition 3.12, this family serves as a universal upper bound for the shortness of any cyclotomic polynomial with root multiplicities bounded by N.

Our approach was initially computational. However the combinatorial explosion of the search space forced us to restrict our computation to short polynomial of degree less or equal than 23. We determined numerical upper bounds for $N \leq 36$. These results highlighted a strong structural heuristic: optimal multisets of cyclotomic factors tend to be "closed" under divisibility, favoring the formation of sparse polynomials of the form $z^m - 1$.

From a theoretical standpoint, we demonstrated that the growth of the shortness is significantly lower than the trivial bound 2^N. We showed that the use of Φ_2 allows us to reach a bound in $O(2^N/\sqrt{N})$. Furthermore, by exploiting connections with Euler's Pentagonal Number Theorem and truncated Euler products, we proved that the asymptotic growth can be bounded via Sudler's constant $K \approx 0.19861$, leading to an exponential improvement over the naive binomial expansion. However, a significant gap remains between these exponential upper bounds and the linear lower bound $(N+1)$ established via polynomial weight arguments.

Applying the uncertainty principle on finite abelian groups allowed us to show the necessity of certain cyclotomic factors within the short polynomial, namely chains of the form $p, \ldots, p^k$ for primes satisfying $p > \mathrm{sh}(\Phi_1^N)$ and p^k appearing in the short polynomial. Furthermore, Proposition 4.8 showed that Φ_2 must be a factor for all $N \geq 3$. These results reinforce the idea that products of the form $z^m - 1$ are the most effective tools for reducing length. This heuristic is summarized in the following conjecture:

Conjecture 7.1. *For all $N \in \mathbb{N}^*$, there exists a short polynomial Q for Φ_1^N such that the product $\Phi_1^N Q$ is of the form:*

$$\Phi_1^N(z)Q(z) = (z^2-1)\prod_{m\in\mathcal{M}}(z^m-1)$$

for some multiset $\mathcal{M}$ of $\mathbb{N}$.

A deeper study of the "Sudler Mountain" and the failure of cancellation in truncated Euler products could lead to the construction of modified polynomials that "smooth out" the peak of central coefficients, potentially yielding tighter upper bounds.

Additionally, a perspective not fully exploited in this thesis involves treating the product of cyclotomic polynomials as a multivariate polynomial:

$$\tilde{P}(z_1, \ldots, z_K) = \prod_{k=1}^{K} \Phi_{n_k}(z_k)$$

In order to apply results similar to those found in the work of Tao [28] (specifically Lemma 1.2). While no practicable result was found using this method during this project, it remains a promising direction.

To conclude, I would like to quote Andrew Wiles:

> Perhaps I could best describe my experience of doing mathematics in terms of entering a dark mansion. You go into the first room and it's dark, completely dark. You stumble around, bumping into the furniture. Gradually, you learn where each piece of furniture is. And finally, after six months or so, you find the light switch and turn it on. Suddenly, it's all illuminated and you can see exactly where you were.

This summarizes my own experience quite well. I stumbled into a lot of furniture, and even found a big couch (my conjecture). However, I have yet to find the light switch, and perhaps the couch was actually a table all along.

Appendix A Additional Figures

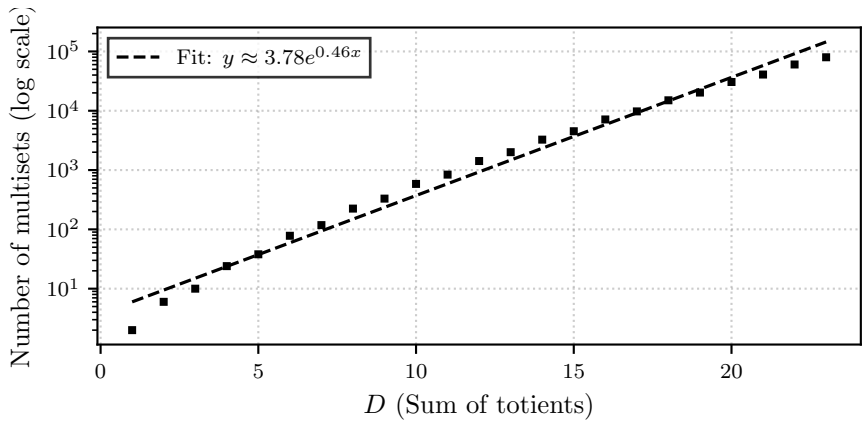

Figure 5: Evolution of the number of multisets in function of the degree D in log scale for the y-axis with an exponential fit. In other words the y-axis represent the number of existing cyclotomic polynomials of degree D.

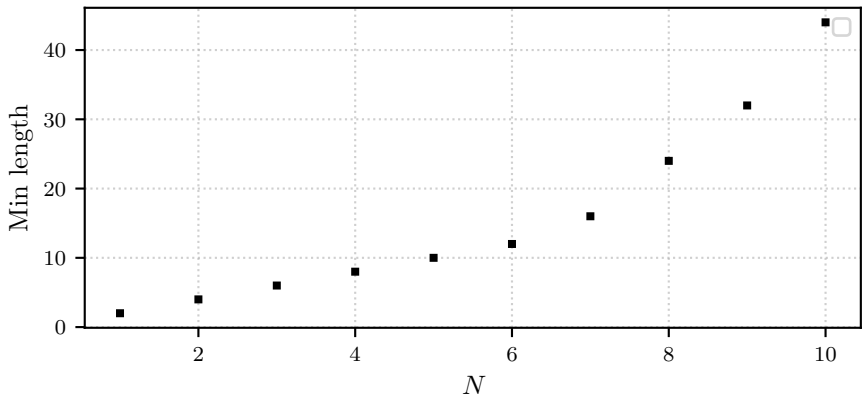

Figure 6: Minimal length of the product $L(\Phi_1^N \prod_{n \in N} \Phi_n)$ found for cyclotomic polynomials of degree up to 23, as a function of $N \in \{1, \ldots, 10\}$.

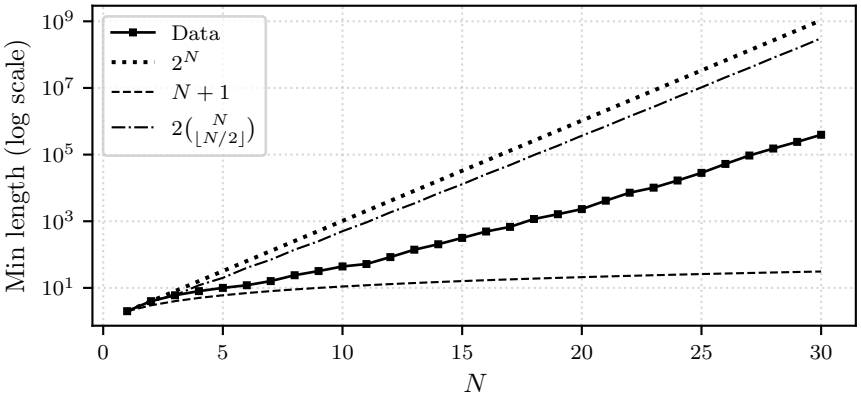

Figure 7: Minimal length of the product $L(\Phi_1^N \prod_{n \in N} \Phi_n)$ found for cyclotomic polynomials of degree up to 30, as a function of $N \in \{1, \ldots, 10\}$. All the theoritical bounds found in section 4.1 are present to compare.

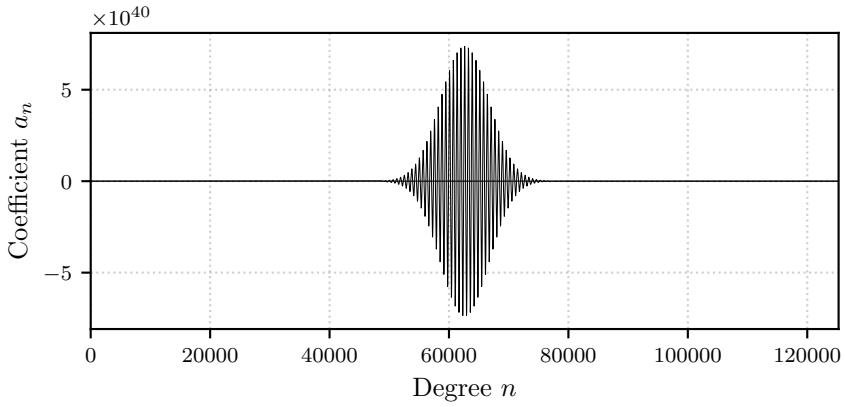

Figure 8: Value of the coefficients of the truncated Euler polynomial Q_{500} in function of the degree n.

Appendix B Additional Tables

This appendix presents detailed numerical results regarding the upper bound of the shortness of Φ_1^N. These values were obtained through the computational search algorithm described in Section 4.

In the table below, the column *Min Length* corresponds to the minimum length $L(\Phi_1^N Q)$ found for a specific N, which serves as an upper bound for $\mathrm{sh}(\Phi_1^N)$. The column *Multiset* lists the indices $\mathcal{N} = \{n_1, n_2, \dots\}$ of the cyclotomic factors composing the multiplier polynomial $Q = \prod_{n \in \mathcal{N}} \Phi_n$. The table stops at $N = 36$, but computations were carried out up to $N = 898$. The trend continues and appears to consist solely of multisets of 2's after $N = 26$. This method becomes less effective as N grows; for example, at $N = 898$, the best polynomial found was Φ_2, which matches the first derived upper bound. The "Closed" column indicates if the multiset contains all divisors (> 1) of its elements. Only one multiset violates this rule: for $N = 4$, a multiset contains 12 without 6.

Table 1: Summary of Multisets by k

N	Min Length	Multiset	Closed?
1	2	$\{2\}$	Yes
1	2	$\{3\}$	Yes
1	2	$\{2, 4\}$	Yes
1	2	$\{5\}$	Yes
1	2	$\{2, 4, 8\}$	Yes
1	2	$\{2, 3, 6\}$	Yes
1	2	$\{7\}$	Yes
1	2	$\{3, 9\}$	Yes
1	2	$\{2, 5, 10\}$	Yes
1	2	$\{11\}$	Yes
1	2	$\{2, 3, 4, 6, 12\}$	Yes
1	2	$\{13\}$	Yes
1	2	$\{2, 7, 14\}$	Yes
1	2	$\{3, 5, 15\}$	Yes
1	2	$\{2, 4, 8, 16\}$	Yes
1	2	$\{17\}$	Yes
1	2	$\{2, 3, 6, 9, 18\}$	Yes
1	2	$\{19\}$	Yes
1	2	$\{2, 4, 5, 10, 20\}$	Yes
1	2	$\{3, 7, 21\}$	Yes
1	2	$\{2, 11, 22\}$	Yes
1	2	$\{23\}$	Yes
1	2	$\{2, 3, 4, 6, 8, 12, 24\}$	Yes
2	4	$\{2, 3\}$	Yes
2	4	$\{2, 4\}$	Yes
2	4	$\{2, 2, 4\}$	Yes
2	4	$\{3, 3\}$	Yes

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Table 1 – continued from previous page

N	Min Length	Multiset	Closed?
2	4	$\{5\}$	Yes
2	4	$\{2,3,4\}$	Yes
2	4	$\{2,3,6\}$	Yes
2	4	$\{2,5\}$	Yes
2	4	$\{2,2,3,6\}$	Yes
2	4	$\{2,2,4,4\}$	Yes
2	4	$\{3,5\}$	Yes
2	4	$\{7\}$	Yes
2	4	$\{2\}$	Yes
2	4	$\{2,2\}$	Yes
2	4	$\{3\}$	Yes
2	4	$\{2,3,3,6\}$	Yes
2	4	$\{2,4,5\}$	Yes
2	4	$\{2,4,8\}$	Yes
2	4	$\{2,7\}$	Yes
2	4	$\{2,2,3,4,6\}$	Yes
2	4	$\{2,2,4,8\}$	Yes
2	4	$\{3,7\}$	Yes
2	4	$\{3,9\}$	Yes
2	4	$\{5,5\}$	Yes
2	4	$\{2,3,4,8\}$	Yes
2	4	$\{2,3,5,6\}$	Yes
2	4	$\{2,3,9\}$	Yes
2	4	$\{2,4,7\}$	Yes
2	4	$\{2,5,10\}$	Yes
2	4	$\{2,2,3,3,6,6\}$	Yes
2	4	$\{2,2,4,4,8\}$	Yes
2	4	$\{2,2,5,10\}$	Yes
2	4	$\{3,3,9\}$	Yes
2	4	$\{5,7\}$	Yes
2	4	$\{11\}$	Yes
2	4	$\{2,3,4,6,12\}$	Yes
2	4	$\{2,3,4,9\}$	Yes
2	4	$\{2,3,5,10\}$	Yes
2	4	$\{2,3,6,7\}$	Yes
2	4	$\{2,4,5,8\}$	Yes
2	4	$\{2,11\}$	Yes
2	4	$\{2,2,3,4,6,8\}$	Yes
2	4	$\{2,2,3,4,6,12\}$	Yes
2	4	$\{2,2,4,5,10\}$	Yes
2	4	$\{3,5,9\}$	Yes
2	4	$\{3,11\}$	Yes
2	4	$\{7,7\}$	Yes
2	4	$\{13\}$	Yes
2	4	$\{2,3,3,4,6,12\}$	Yes
2	4	$\{2,3,3,6,9\}$	Yes

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Table 1 – continued from previous page

N	Min Length	Multiset	Closed?
2	4	$\{2, 4, 7, 8\}$	Yes
2	4	$\{2, 4, 11\}$	Yes
2	4	$\{2, 5, 5, 10\}$	Yes
2	4	$\{2, 7, 14\}$	Yes
2	4	$\{2, 13\}$	Yes
2	4	$\{2, 2, 3, 4, 4, 6, 12\}$	Yes
2	4	$\{2, 2, 3, 5, 6, 10\}$	Yes
2	4	$\{2, 2, 4, 4, 8, 8\}$	Yes
2	4	$\{2, 2, 7, 14\}$	Yes
2	4	$\{3, 5, 15\}$	Yes
2	4	$\{3, 7, 9\}$	Yes
2	4	$\{3, 13\}$	Yes
2	4	$\{5, 11\}$	Yes
2	4	$\{2, 3, 4, 5, 6, 12\}$	Yes
2	4	$\{2, 3, 4, 8, 9\}$	Yes
2	4	$\{2, 3, 5, 15\}$	Yes
2	4	$\{2, 3, 6, 11\}$	Yes
2	4	$\{2, 3, 7, 14\}$	Yes
2	4	$\{2, 4, 8, 16\}$	Yes
2	4	$\{2, 4, 13\}$	Yes
2	4	$\{2, 5, 7, 10\}$	Yes
2	4	$\{2, 2, 3, 3, 4, 6, 6, 12\}$	Yes
2	4	$\{2, 2, 4, 5, 8, 10\}$	Yes
2	4	$\{2, 2, 4, 7, 14\}$	Yes
2	4	$\{2, 2, 4, 8, 16\}$	Yes
2	4	$\{3, 3, 5, 15\}$	Yes
2	4	$\{3, 3, 9, 9\}$	Yes
2	4	$\{5, 13\}$	Yes
2	4	$\{7, 11\}$	Yes
2	4	$\{17\}$	Yes
2	4	$\{2, 3, 4, 5, 15\}$	Yes
2	4	$\{2, 3, 4, 6, 7, 12\}$	Yes
2	4	$\{2, 3, 4, 8, 16\}$	Yes
2	4	$\{2, 3, 5, 9, 10\}$	Yes
2	4	$\{2, 3, 6, 9, 18\}$	Yes
2	4	$\{2, 3, 6, 13\}$	Yes
2	4	$\{2, 4, 8, 11\}$	Yes
2	4	$\{2, 5, 7, 14\}$	Yes
2	4	$\{2, 17\}$	Yes
2	4	$\{2, 2, 3, 4, 4, 6, 8, 12\}$	Yes
2	4	$\{2, 2, 3, 6, 7, 14\}$	Yes
2	4	$\{2, 2, 3, 6, 9, 18\}$	Yes
2	4	$\{2, 2, 4, 4, 8, 16\}$	Yes
2	4	$\{2, 2, 5, 5, 10, 10\}$	Yes
2	4	$\{3, 5, 5, 15\}$	Yes
2	4	$\{3, 9, 11\}$	Yes

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N	Min Length	Multiset	Closed?
2	4	$\{3, 17\}$	Yes
2	4	$\{7, 13\}$	Yes
2	4	$\{19\}$	Yes
2	4	$\{2, 3, 3, 4, 6, 9, 12\}$	Yes
2	4	$\{2, 3, 3, 5, 6, 15\}$	Yes
2	4	$\{2, 3, 3, 6, 9, 18\}$	Yes
2	4	$\{2, 4, 5, 8, 16\}$	Yes
2	4	$\{2, 4, 5, 10, 20\}$	Yes
2	4	$\{2, 4, 8, 13\}$	Yes
2	4	$\{2, 4, 17\}$	Yes
2	4	$\{2, 5, 10, 11\}$	Yes
2	4	$\{2, 7, 7, 14\}$	Yes
2	4	$\{2, 19\}$	Yes
2	4	$\{2, 2, 3, 4, 5, 6, 10, 12\}$	Yes
2	4	$\{2, 2, 3, 4, 6, 8, 16\}$	Yes
2	4	$\{2, 2, 3, 4, 6, 9, 18\}$	Yes
2	4	$\{2, 2, 4, 5, 10, 20\}$	Yes
2	4	$\{2, 2, 4, 7, 8, 14\}$	Yes
2	4	$\{3, 5, 7, 15\}$	Yes
2	4	$\{3, 7, 21\}$	Yes
2	4	$\{3, 9, 13\}$	Yes
2	4	$\{3, 19\}$	Yes
2	4	$\{5, 17\}$	Yes
2	4	$\{11, 11\}$	Yes
2	4	$\{2, 3, 4, 5, 8, 15\}$	Yes
2	4	$\{2, 3, 4, 5, 10, 20\}$	Yes
2	4	$\{2, 3, 4, 6, 11, 12\}$	Yes
2	4	$\{2, 3, 5, 6, 9, 18\}$	Yes
2	4	$\{2, 3, 6, 17\}$	Yes
2	4	$\{2, 3, 7, 9, 14\}$	Yes
2	4	$\{2, 3, 7, 21\}$	Yes
2	4	$\{2, 4, 7, 8, 16\}$	Yes
2	4	$\{2, 4, 19\}$	Yes
2	4	$\{2, 5, 10, 13\}$	Yes
2	4	$\{2, 11, 22\}$	Yes
2	4	$\{2, 2, 3, 3, 4, 4, 6, 6, 12, 12\}$	Yes
2	4	$\{2, 2, 3, 3, 6, 6, 9, 18\}$	Yes
2	4	$\{2, 2, 4, 4, 5, 10, 20\}$	Yes
2	4	$\{2, 2, 4, 4, 8, 8, 16\}$	Yes
2	4	$\{2, 2, 5, 7, 10, 14\}$	Yes
2	4	$\{2, 2, 11, 22\}$	Yes
2	4	$\{3, 3, 5, 9, 15\}$	Yes
2	4	$\{3, 3, 7, 21\}$	Yes
2	4	$\{5, 19\}$	Yes
2	4	$\{7, 17\}$	Yes
2	4	$\{11, 13\}$	Yes

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Table 1 – continued from previous page

N	Min Length	Multiset	Closed?
2	4	$\{23\}$	Yes
2	4	$\{2,3,4,6,8,12,24\}$	Yes
2	4	$\{2,3,4,6,12,13\}$	Yes
2	4	$\{2,3,4,7,21\}$	Yes
2	4	$\{2,3,4,8,9,16\}$	Yes
2	4	$\{2,3,5,5,10,15\}$	Yes
2	4	$\{2,3,6,7,9,18\}$	Yes
2	4	$\{2,3,6,19\}$	Yes
2	4	$\{2,3,11,22\}$	Yes
2	4	$\{2,4,5,5,10,20\}$	Yes
2	4	$\{2,4,8,17\}$	Yes
2	4	$\{2,7,11,14\}$	Yes
2	4	$\{2,23\}$	Yes
3	6	$\{2\}$	Yes
3	6	$\{2,3\}$	Yes
3	6	$\{2,2,2,4\}$	Yes
3	6	$\{2,3,4\}$	Yes
3	6	$\{2,3,5\}$	Yes
3	6	$\{2,4,5\}$	Yes
3	6	$\{2,2,2,3,4,6\}$	Yes
3	6	$\{2,3,3,3,6\}$	Yes
3	6	$\{2,3,5,6\}$	Yes
3	6	$\{2,3,4,7\}$	Yes
3	6	$\{2,3,6,7\}$	Yes
3	6	$\{2,5,7\}$	Yes
3	6	$\{2,2,2,3,4,6,8\}$	Yes
3	6	$\{2,2,2,4,4,4,8\}$	Yes
3	6	$\{2,3,4,5,8\}$	Yes
3	6	$\{2,4,7,8\}$	Yes
3	6	$\{2,3,3,3,6,9\}$	Yes
3	6	$\{2,3,4,5,9\}$	Yes
3	6	$\{2,3,4,8,9\}$	Yes
3	6	$\{2,3,7,9\}$	Yes
3	6	$\{2,2,2,3,4,5,6,10\}$	Yes
3	6	$\{2,2,2,4,5,8,10\}$	Yes
3	6	$\{2,3,5,7,10\}$	Yes
3	6	$\{2,3,5,9,10\}$	Yes
3	6	$\{2,5,5,5,10\}$	Yes
3	6	$\{2,3,4,8,11\}$	Yes
3	6	$\{2,3,5,6,11\}$	Yes
3	6	$\{2,3,9,11\}$	Yes
3	6	$\{2,4,7,11\}$	Yes
3	6	$\{2,5,10,11\}$	Yes
3	6	$\{2,2,2,3,3,3,4,6,6,6,12\}$	Yes
3	6	$\{2,2,2,3,4,4,4,6,8,12\}$	Yes

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Table 1 – continued from previous page

N	Min Length	Multiset	Closed?
3	6	$\{2, 2, 2, 3, 4, 5, 6, 10, 12\}$	Yes
3	6	$\{2, 3, 3, 3, 4, 6, 9, 12\}$	Yes
3	6	$\{2, 3, 4, 5, 6, 7, 12\}$	Yes
3	6	$\{2, 3, 4, 6, 11, 12\}$	Yes
3	6	$\{2, 3, 4, 6, 12, 13\}$	Yes
3	6	$\{2, 3, 4, 9, 13\}$	Yes
3	6	$\{2, 3, 5, 10, 13\}$	Yes
3	6	$\{2, 3, 6, 7, 13\}$	Yes
3	6	$\{2, 4, 5, 8, 13\}$	Yes
3	6	$\{2, 11, 13\}$	Yes
4	8	$\{2, 3\}$	Yes
4	8	$\{2, 2, 3, 4\}$	Yes
4	8	$\{2, 3, 5\}$	Yes
4	8	$\{2, 3, 4, 5\}$	Yes
4	8	$\{2, 2, 2, 2, 3, 4, 6\}$	Yes
4	8	$\{2, 3, 4, 7\}$	Yes
4	8	$\{2, 2, 3, 4, 5, 6\}$	Yes
4	8	$\{2, 3, 5, 7\}$	Yes
4	8	$\{2, 2, 3, 4, 5, 8\}$	Yes
4	8	$\{2, 3, 4, 5, 9\}$	Yes
4	8	$\{2, 3, 5, 6, 7\}$	Yes
4	8	$\{2, 2, 2, 2, 3, 4, 4, 6, 8\}$	Yes
4	8	$\{2, 3, 3, 3, 3, 6, 9\}$	Yes
4	8	$\{2, 2, 2, 2, 3, 4, 5, 6, 10\}$	Yes
4	8	$\{2, 2, 3, 4, 6, 7, 8\}$	Yes
4	8	$\{2, 3, 4, 5, 7, 12\}$	No
4	8	$\{2, 3, 5, 6, 11\}$	Yes
4	8	$\{2, 3, 5, 7, 9\}$	Yes
4	8	$\{2, 2, 3, 4, 5, 7, 10\}$	Yes
4	8	$\{2, 3, 4, 7, 8, 9\}$	Yes
4	8	$\{2, 3, 4, 7, 11\}$	Yes
4	8	$\{2, 2, 2, 2, 3, 4, 5, 6, 8, 10\}$	Yes
4	8	$\{2, 2, 3, 4, 5, 6, 7, 12\}$	Yes
4	8	$\{2, 3, 4, 5, 8, 11\}$	Yes
4	8	$\{2, 3, 6, 7, 13\}$	Yes
5	10	$\{2, 3, 5, 7\}$	Yes
5	10	$\{2, 3, 4, 5, 7\}$	Yes
6	12	$\{2, 2, 3, 4, 5\}$	Yes
6	12	$\{2, 2, 3, 4, 5, 7\}$	Yes
6	12	$\{2, 2, 3, 4, 5, 7, 8\}$	Yes
6	12	$\{2, 2, 3, 3, 3, 4, 5, 6, 9\}$	Yes
7	16	$\{2, 2, 3, 4, 5, 7\}$	Yes
8	24	$\{2, 2, 2, 3, 4, 5, 7\}$	Yes

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N	Min Length	Multiset	Closed?
8	24	$\{2,2,3,3,4,5,7\}$	Yes
8	24	$\{2,2,3,4,5,5,7\}$	Yes
8	24	$\{2,2,2,3,3,4,5,6,7\}$	Yes
8	24	$\{2,2,2,3,4,4,5,7,8\}$	Yes
9	32	$\{2,2,2,3,3,3,4,5,6,7\}$	Yes
10	44	$\{2,2,2,3,3,4,5,7\}$	Yes
10	44	$\{2,2,2,3,3,4,5,5,7\}$	Yes
11	52	$\{2,2,2,3,3,4,5,5,7\}$	Yes
12	84	$\{2,2,2,3,3,4,5,5,7\}$	Yes
13	140	$\{2,2,2,3,3,4,5,5,7\}$	Yes
14	204	$\{2,2,2,2,3,3,3,4,5,7\}$	Yes
15	316	$\{2,2,2,2,2,3,3,3,4,4,5,5\}$	Yes
16	492	$\{2,2,2,2,3,3,3,4,5,7\}$	Yes
17	678	$\{2,2,2,2,2,3,3,3,4,5,7\}$	Yes
18	1168	$\{2,2,2,2,2,3,3,3,4,5,7\}$	Yes
19	1626	$\{2,2,2,2,2,2,2,3,3,3,3,4,4,5\}$	Yes
20	2312	$\{2,2,2,2,2,2,2,3,3,3,3,4,4,5\}$	Yes
21	4144	$\{2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
22	7188	$\{2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
23	10186	$\{2,2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
24	16692	$\{2,2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
25	28284	$\{2,2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
26	52480	$\{2,2,2,2,2,2,2,3,3,3,3,3,4,5\}$	Yes
27	93820	$\{2,2,2,2,2,2,2,2,3,3,3,3,4,5\}$	Yes
28	152216	$\{2,2,2,2,2,2,2,2,3,3,3,3,4,5\}$	Yes
29	240968	$\{2,2,2,2,2,2,2,2,2,3,3,3,3,3,3,4\}$	Yes
30	393544	$\{2,2,2,2,2,2,2,2,2,3,3,3,3,3,3,4\}$	Yes
31	12326972	$\{2,2,2,2,2,2,2,2,3\}$	Yes
32	22699700	$\{2,2,2,2,2,2,2,2,3\}$	Yes
33	40690752	$\{2,2,2,2,2,2,2,2,3\}$	Yes
34	76203660	$\{2,2,2,2,2,2,2,2,3\}$	Yes
35	137116816	$\{2,2,2,2,2,2,2,2,3\}$	Yes
36	260401260	$\{2,2,2,2,2,2,2,2,3\}$	Yes

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